

BICYCLES AND MOTORCYCLES



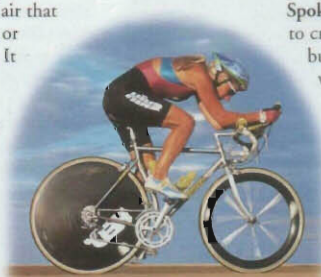
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FUN, AND ENVIRONMENTALLY friendly, the bicycle is the simplest form of mechanical transport. A bicycle, or bike, is a two-wheeled machine that converts human energy into propulsion; a motorcycle, or motorbike, is a bicycle with an engine. Modern motorcycles are complex, with engine sizes ranging from 50cc (cubic centimetres) to more than 1,000cc. In many countries, such as China, most people travel or transport goods by bicycle. Across the world, bicycles and motorcycles are used for sport and leisure.

Reducing drag

Drag is the resistance of air that can slow down a bicycle or motorcycle and its rider. It is reduced by creating a streamlined shape for the air to flow around – some competitive bicycle riders even shave their legs to achieve this streamlined effect.

Time-trial bike



Cannondale SH600, hybrid

Spokes are arranged to create a strong but lightweight wheel.

Tyres fitted on a metal wheel rim give a smooth, quiet ride over small bumps; mountain bikes have fatter tyres to handle rough and rocky terrain.

Parts of a bicycle

From a mountain bike to a racing bike, or a hybrid (a cross between the two), all bicycles are built in a similar way. Designed to be easy to pedal and comfortable, the weight is also important, as it affects the speed at which the bike can be propelled.

Saddles are adjustable, moving up and down to accommodate different riders.

Gears, operated by levers, move the chain between different-sized gear wheels, to change the speed at which the wheels turn.

Handlebars may be dropped for riding crouched.

Brakes are controlled by pulling levers on handlebars, which force brake blocks against wheel rims to slow the bicycle down.

Seat post slides in and out of frame to adjust seat level.

Frame, made from metal tubes, to support the rider.

Brake cable

Chain wheel

Pedals, attached to the chain wheel, are pushed to turn the wheel.

Wheel hub secures the wheel to frame.



Parts of a motorcycle

Like a bicycle, a motorcycle has a frame, a rear wheel that drives it along, a front wheel for steering, and controls on the handlebars. Like a car, it has an internal combustion engine and suspension. The suspension supports the motorcycle's body on the wheels, and stops it being affected by the bumping of the wheels on the road.

Two-stroke engine with one cylinder. Larger motorcycles have more cylinders.

Lightweight frame

Fuel tank

Speedometer

Indicator and warning lights

Ignition switch

Engine rev counter



Motorcycle instrument panel

Motorcycles have an instrument panel in the centre of the handlebars. Control switches for lights and indicators can be operated with hands on the handlebars.

Front suspension



1992 Yamaha FZR1000 Exup

Three-spoke alloy rear wheel, supported by suspension strut.

Motorcycle tyres grip the road even when the motorcycle leans over at corners. These are smooth, treadless, "slick" racing tyres.

Riding a motorcycle

A motorcycle rider changes speed by twisting the right-hand handlebar grip, and changes gear by flicking a foot lever up or down. The front brakes are operated by hand, and the rear brakes by foot. To go round a corner, the rider turns the handlebars and leans the motorcycle over.



Small engine for speed and economy

Open "step in" frame



SFX moped

Mopeds and scooters

Small motorcycles used for short journeys in towns and cities are called mopeds or scooters. They have small engines, so they cannot go very fast, but are very economical. Mopeds, restricted to a 50cc engine, have pedals which the rider can use on steep hills.

Timeline

1839 Kirkpatrick MacMillan, a Scot, invents a lever-driven bicycle.

1863 The French Michaux brothers build the first pedal-powered bike, a velocipede.

1868 The Michaux brothers add a steam engine to a bike, creating the first motorcycle.

1885 In England, James Starley makes modern-style bicycles.

1885 German Gottlieb Daimler builds an engine-powered tricycle (below).



1901 The 1901 Werner is the first practical road-going motorcycle.

1914–18 Motorcycles used extensively in World War I.

1963 Dutchman Van Wijnen designs what will become the Ecocar – covered pedal-powered transport.

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